

Auditing Image Use in Medical Vision Language Models

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Abstract

Medical vision-language models can obtain strong classification scores from radiology reports even when their corresponding images contribute little to the decision. This risk is greatest when evaluation labels and text inputs share report provenance. We audited five fusion conditions on 2,953 frontal MIMIC-CXR studies divided at patient level into 2,049 training, 449 validation, and 455 test studies. The primary intervention replaced each radiograph with a constant gray image while preserving its report, target, architecture, and evaluation cohort. The Image Contribution Metric (ICM) measured the relative Macro-F1 change between the full model and this report-preserving control. Across eight baseline seeds, Macro-F1 was 0.9397 ± 0.0061 , whereas the retained Gray-Square control achieved 0.9381. Baseline ICM was 0.0016 ± 0.0065 with a 95% bootstrap confidence interval of $[-0.0026, 0.0061]$. A signed-rank test did not reject zero ($p = 0.4219$), and two one-sided tests supported equivalence inside the operational interval $[-0.01, 0.01]$ ($p = 0.0040$). None of the tested interventions produced a robust increase in incremental image contribution. The product-of-experts condition increased expected calibration error from 0.0237 to 0.0621, while retained baseline gates stayed close to their initial values, with an overall mean of 0.49975. These findings show that high multimodal performance can coexist with negligible change after image replacement. ICM, matched interventions, calibration, and gate inspection provide complementary checks for this form of benchmark dependence. The conclusion is limited to the report-linked cohort and retained implementation and does not imply that chest radiographs lack diagnostic value.

Keywords: medical vision-language model; label circularity; shortcut learning; modality audit; Image Contribution Metric; equivalence testing; MIMIC-CXR.

1 Introduction

Chest radiography is interpreted in a clinical information environment rather than as an isolated image. Radiologists draw on image appearance, indication, prior examinations, and other contextual information before expressing their assessment in a report. Medical vision-language models (VLMs) mirror part of that workflow by combining a radiograph with its report or another clinical text source. Large paired resources such as MIMIC-CXR make this combination possible at scale and have supported classification, representation learning, report generation, and question answering [1]. The appeal is clear: an image can carry anatomical and pathological evidence while text can represent symptoms, uncertainty, and prior interpretation.

The success of paired pretraining has strengthened this view. ConVIRT learns visual representations from image-text agreement, GLoRIA models global and local correspondence, and BioViL emphasizes biomedical report semantics [2–4]. Clinical language encoders further improve the representation of medical terminology, negation, and contextual phrases [5, 6]. These developments establish that reports are useful supervision and useful inputs.

They do not establish that both modalities affect a particular downstream prediction. A model can be multimodal by construction while behaving like a unimodal model on the evaluated endpoint.

This distinction matters because predictive performance is a property of a model, a cohort, and a target definition. It is not direct evidence about the reason for a prediction. Deep networks often exploit shortcuts that remain stable within a benchmark but fail to represent the intended clinical construct [7]. In radiology, scanner characteristics, institutional practice, population composition, acquisition protocol, and annotation procedure can all become predictive signals [8]. In a VLM, the report can form an additional shortcut when it encodes most of the information rewarded by the label. The image branch may then add little measurable information even though the architecture contains a large visual encoder, cross-attention, and an explicit fusion gate.

Recent evidence shows why this possibility cannot be dismissed as an edge case. Multimodal medical systems can return correct answers with flawed visual rationales, and several chest-radiography models retain substantial performance when visual information is removed or replaced [9, 10]. Clinical context can materially change measured per-

formance, so a headline score may partly describe the informativeness of the supplied context rather than visual recognition [11]. Audit frameworks have consequently begun to separate answer accuracy from modality dependence and to identify settings in which apparently successful models occupy a dangerous combination of high performance and weak visual grounding [12]. The common implication is that image use should be measured through intervention rather than inferred from model components or explanations.

Target provenance creates a particularly direct threat in MIMIC-CXR-derived classification. CheXpert converts report language into uncertainty-aware pathology labels [13]. When the predictor also receives the same report, or its **findings** section, the target and the text input share a source. The resulting task may remain useful for studying report-linked prediction, fusion behaviour, or computational auditing. It no longer isolates independent recognition from the radiograph. A model can be rewarded for recovering the linguistic evidence that contributed to label generation. Errors or subgroup effects in natural-language labelers can also propagate into the training target and the apparent model performance [14, 15].

This form of construct overlap complicates the interpretation of debiasing. Methods developed for visual question answering often expose, down-weight, or combine an easy language-only expert with a multimodal expert [16–18]. Those methods are well motivated when the language pathway captures a nuisance prior. In a report-derived classification task, however, the language pathway can be close to the mechanism that produced the reference label. Penalizing it may lower the score or distort confidence without generating additional visual information. The result cannot be interpreted as better image use unless an image-preserving comparison demonstrates that change. Prototype and counterfactual methods can improve transparency, but explanation quality and causal input dependence remain separate questions [19].

Three gaps motivate the present study. First, modality controls are often not matched closely enough to the full model. Removing the report, training a new unimodal architecture, or using a different cohort changes more than the availability of image information. A report-preserving image intervention provides a cleaner contrast because the report, target, evaluation cases, and multimodal path remain fixed. Second, fusion parameters and saliency maps are frequently read as evidence of reliance. A balanced gate can remain close to its

initial value, and an attention or activation map can look plausible even when the prediction barely changes after image replacement [20, 21]. Third, a non-significant difference does not demonstrate negligible effect. Equivalence testing is needed when the scientific question concerns whether an effect lies inside a prespecified operational margin [22].

We therefore conduct a controlled modality audit on a three-class MIMIC-CXR-derived task comprising Normal, Pneumonia, and Pneumothorax. The baseline combines a DenseNet-121 image encoder and Bio_ClinicalBERT report encoder through cross-attention and a bounded per-class gate. Four additional conditions test a learned text-prior product-of-experts design, a noise-sensitivity (anti-consistency) objective, an exploratory grounding-aware condition, and adaptive gating. The primary intervention replaces every radiograph with a constant gray image while preserving the report. We define the Image Contribution Metric (ICM) as the relative Macro-F1 difference between the full model and this Gray-Square reference, then pair it with seed-level uncertainty and two one-sided equivalence tests.

The analysis is organized around three questions. Does a real radiograph change performance when the report remains available? Do the tested fusion and debiasing conditions increase that incremental contribution? Do retained gate values and qualitative saliency provide evidence consistent with the behavioural intervention? Answering these questions requires performance, calibration, paired testing, per-class error analysis, and provenance to be read together. The contribution is therefore an audit procedure and an empirical case study of a report-linked benchmark. It is not a new diagnostic model, and it does not imply that radiographs are unnecessary in clinical care.

2 Related work

This section places the audit between three research areas: medical image-text representation learning, shortcut mitigation, and intervention-based model evaluation. Table 1 summarizes the closest methods and identifies the evidence each contributes. The comparison is conceptual; reported models use different datasets, targets, and evaluation designs and their numerical results are not directly comparable.

2.1 Medical image-text learning

Large image-report collections allow supervision to be obtained from text that already exists in clinical practice. ConVIRT uses paired contrastive

learning to align images and reports, GLoRIA combines global and local representations, and BioViL strengthens the use of biomedical language semantics [2–4]. These methods show that text can improve visual representations, especially when manual image labels are scarce. They also illustrate why the evaluation target must be kept distinct from the supervision source. A representation can encode useful image content while a downstream classifier still relies mainly on report-derived features.

Report encoders add another source of asymmetry. BERT and clinical-domain variants encode lexical and contextual cues at a level that a relatively small image branch may not match [5, 6]. In radiology reports, explicit mentions, negation, and uncertainty can be closely aligned with the output classes. If those same phrases were used to produce the labels, a high score is expected from the language pathway. It is therefore necessary to distinguish the benefit of text for prediction from evidence that the image changed the prediction.

2.2 Shortcut learning and language priors

Shortcut learning provides the broader framework for this distinction. Predictive systems may use features that work in a development distribution but do not represent the intended task [7]. Radiology models are exposed to many such features, including acquisition artifacts and institution-specific correlations [8]. A multimodal model adds a more direct shortcut when one modality contains a near-explicit description of the target.

Language-prior research in visual question answering offers several mitigation strategies. VQA-CP changes train-test answer priors, ensemble approaches model the easy route, and RUBi or product-of-experts losses reduce the influence of a language-only component [16–18]. These designs assume that the language signal is predictive but misleading. That assumption is weaker in report-linked radiology classification. Here the report can be both clinically meaningful and structurally connected to the label. Suppression may remove rewarded information, but it cannot create an independent image reference standard.

2.3 Circularity and modality audits

Studies of medical VLMs increasingly test whether output quality reflects visual evidence. Jin et al. separate correct answers from the quality of their multimodal rationale, while Wang et al. show that the

usefulness of context can alter chest-radiography performance [9, 11]. Lotfinia et al. use image removal and related interventions across multiple systems, demonstrating that text-only or visually disrupted conditions can remain competitive [10]. Sadanandan and Behzadan organize such behaviour into an audit taxonomy and motivate grounding-aware objectives [12]. CXR-NeXus represents a complementary direction based on prototypes and counterfactual consistency [19].

These contributions shift attention from architecture labels to observable behaviour. They also leave several design choices open. A control can be trained separately or applied to a retained model; it can remove, replace, or permute an input; and it can preserve or change the text pathway. Each choice answers a different question. Our Gray-Square condition is deliberately narrow: it asks how much Macro-F1 changes when image content is removed while the report-linked input and the multimodal processing path remain available.

2.4 Fusion parameters and explanations

Gated multimodal units offer an interpretable-looking mechanism for balancing modalities [20]. The numerical gate is still a learned parameter rather than a causal contribution estimate. It can remain near initialization when signals are redundant, gradients are weak, or the loss is already minimized through another route. Attention and Grad-CAM face a related interpretive boundary. They can identify where a model produced high internal activation, but a plausible map does not show how the output would change if that evidence were absent [21]. Reasoning-oriented and prototype-based systems may make intermediate representations easier to inspect, yet their explanations remain complementary to controlled ablation [19, 23].

2.5 Statistical evidence for a negligible effect

Modality audits often report a small difference followed by a conventional test against zero. A non-significant result is ambiguous because it can reflect either a small effect or insufficient precision. Equivalence testing reverses the question by asking whether the effect and its uncertainty fit inside a prespecified interval [22]. This approach is useful for ICM because the scientific claim concerns an operationally negligible change, not exact mathematical equality. The margin must nevertheless be treated as a study

definition rather than a universal clinical threshold.

The comparison also reveals a difference between representation-learning and audit objectives. Representation-learning papers usually ask whether paired text improves transfer, label efficiency, or downstream accuracy. An audit asks whether the resulting predictor depends on the modality named in its claim. Both questions are legitimate, but the first cannot answer the second without an intervention. This is why a model that benefits from image-text pretraining can still show little incremental image contribution on a later report-linked classification task.

The same distinction applies to explanations. A rationale, prototype, or activation map describes an internal or generated representation. A counterfactual input comparison measures output sensitivity. Agreement between the two strengthens an interpretation, while disagreement should be reported rather than reconciled by assuming that the explanation is causal. The current study gives priority to the Gray-Square comparison and treats Grad-CAM as a limited qualitative check. This ordering follows the information available in the retained outputs and avoids using saliency to compensate for a missing behavioural effect.

Finally, prior work differs in the source of uncertainty it reports. Large cross-model audits provide breadth across architectures, whereas repeated seeds provide variation within an implementation. Subject-level paired tests assess whether the same cases change, while equivalence tests address whether the seed-level contribution lies inside an operational interval. The present design does not replace broad external evaluation; it adds depth around a single audited pipeline. Its main value is the explicit connection between target provenance, image replacement, model-level replication, and statistical claim.

Taken together, prior work establishes the plausibility of text dominance and the need for controlled interventions. The present study joins four evidence types that are usually reported separately: a report-preserving image intervention, a relative per-seed contribution measure, equivalence testing, and inspection of retained fusion gates. The combined design is intended to make the claim boundary visible: it evaluates image contribution to one report-linked endpoint rather than visual competence in general.

3 Methods

The methods follow the sequence in Fig. 1: define the report-linked cohort, encode both modalities, compare controlled fusion conditions, and evaluate every retained seed on the same test set. The primary estimand is the incremental contribution of image content when report information remains available. Discrimination, calibration, paired errors, and gate behaviour provide supporting evidence but do not replace the intervention-based comparison.

3.1 Study design, data source, and cohort

We performed a retrospective computational audit. The estimand was incremental image contribution when the report remained available. The principal intervention replaced each radiograph by a constant gray tensor while retaining the report, target, fusion path, and evaluation cohort. Full multimodal outputs were compared with this report-preserving control and with descriptive text-only and image-only controls. Figure 1 presents the image pathway, report pathway, fusion design, and seed-level audit. The visual is a conceptual overview; the retained implementation is specified below.

Panel (a) summarizes DenseNet-121 image encoding, panel (b) shows the Bio.ClinicalBERT report pathway, and panel (c) links five eight-seed conditions to ICM, statistical testing, gate extraction, and retained controls. Panel identifiers are interpreted through this below-figure description rather than as additional method claims. The diagram is not a layer-by-layer computational graph: the retained model uses 512-dimensional projections, findings-only text, separate class logits, and a static baseline gate. DeLong and McNemar use retained seed-42 outputs, and each unimodal or Gray-Square control has one retained run.

MIMIC-CXR JPG images and reports were accessed through official PhysioNet credentialing under the applicable Data Use Agreement [1]. The source flow comprised 377,110 images in 227,835 studies; 227,827 studies had CheXpert labels and 108,626 met the frontal-view eligibility criteria. An analysis cap selected up to 2,500 Normal, 2,500 Pneumonia, and 1,000 Pneumothorax candidates. Of 2,962 attempted downloads, nine JPG files were unavailable, leaving 2,953 usable studies. The inherited patient-level GroupShuffleSplit assigned 2,049 studies to training, 449 to validation, and 455 to test, with no subject overlap. The test set contained 240 Normal, 168 Pneumonia, and 47 Pneumothorax

Table 1. Related methods and the evidence they provide.

Work	Primary setting	Relevant method	Relation to this audit
ConVIRT [2]	Paired medical images and text	Contrastive representation learning	Establishes the value of report supervision but does not isolate image contribution in a report-linked classifier.
GLoRIA [3]	Label-efficient image recognition	Global-local image-text alignment	Motivates cross-modal representation learning; alignment scores are not behavioural evidence of modality dependence.
BioViL [4]	Biomedical vision-language processing	Domain-specific text semantics	Supports the report encoder choice and illustrates the strength of the linguistic pathway.
VQA debiasing [16–18]	Language priors in visual question answering	Prior balancing and product-of-experts training	Provides the LMH rationale, but its nuisance-prior assumption may not hold when reports generate labels.
Context audit [11]	Chest-radiography models with clinical context	Comparison across context conditions	Shows that context changes measured performance and motivates explicit provenance analysis.
Image-removal audit [10]	Medical VLMs at multiple scales	Occlusion, replacement, and modality controls	Closest behavioural precedent; the present work adds relative ICM, equivalence testing, calibration, and gate extraction.
Grounding audit [12]	Multimodal medical AI	Audit taxonomy and grounding-aware objective	Motivates exploratory GACR-B and the separation of high performance from safe modality behaviour.
CXR-NeXus [19]	Interpretable chest X-ray classification	Prototype memory and counterfactual consistency	Provides explanation-oriented evidence; the present intervention focuses on output sensitivity to image replacement.

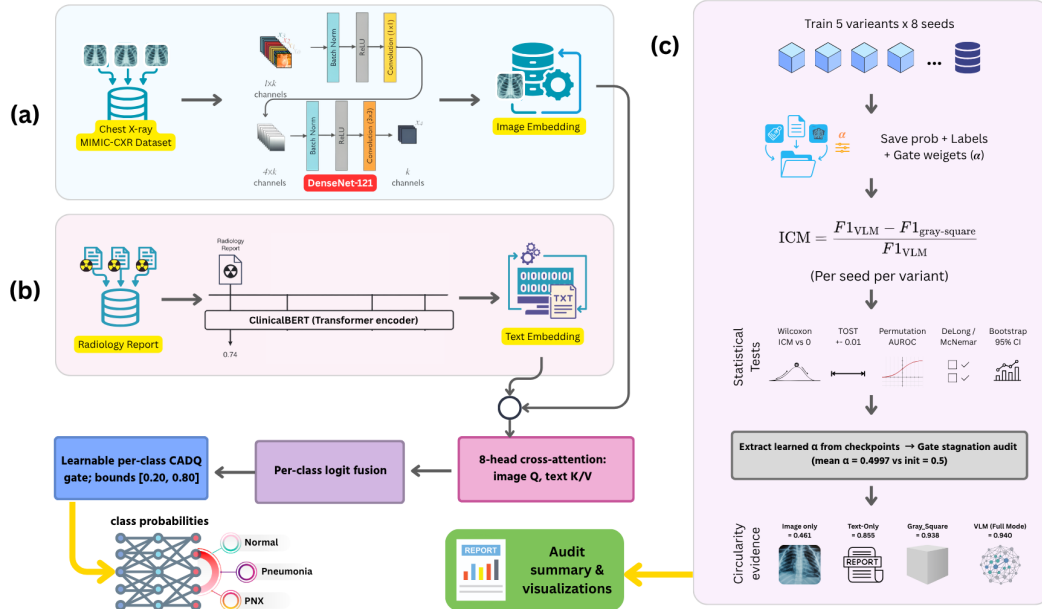


Fig. 1. Image and report encoding, gated fusion, and seed-level modality audit.

studies.

The source counts were retained as a cohort ledger. The sequence was 377,110 images, 227,835 studies,

227,827 labelled studies, 108,626 eligible frontal-view studies, 6,000 capped candidates, 2,962 download attempts, and 2,953 available studies. This account-

ing separates source-database size, clinical eligibility, the practical analysis cap, and file availability. The final class proportions should therefore be understood as properties of the constructed audit cohort rather than prevalence estimates for MIMIC-CXR.

CheXpert generated the targets from report language using the project’s uncertain-to-negative mapping [13]; the model received the same report’s **findings** field. The experiment consequently audits label-report circularity rather than an independent radiograph-only diagnostic task. The replication unit for seed-level inference was a trained model at a fixed seed. Baseline CADQ, LearnedMixinFusion (LMH), noise-sensitivity (anti-consistency), exploratory GACR-B, and Adaptive-CADQ each had eight retained seeds: 42, 123, 456, 789, 1010, 1111, 1212, and 1313. The eight-seed plan covered baseline, LMH, and noise-sensitivity (anti-consistency); Adaptive-CADQ was extended from three to eight seeds after the initial run, and exploratory GACR-B was added post hoc. The latter two analyses are therefore exploratory.

Each study contributed one target to the three-class task. The patient-level split was fixed before model evaluation so that images or reports from the same person could not occur in more than one partition. This choice limits leakage from repeated examinations that may share anatomy, wording patterns, or clinical history. The same 455 test studies were reused across seeds and conditions. This supports paired comparison, provided that repeated subjects are not treated as independent observations when seed-level uncertainty is estimated.

3.2 Audit estimand and evidence hierarchy

The audit is organized around a contrast rather than a model leaderboard. For seed s , the observed full-model Macro-F1 is compared with the retained Gray-Square Macro-F1 after spatial image information has been removed. The report, target definition, class set, and held-out studies remain unchanged. This contrast estimates how much the evaluated endpoint changes when the declared image content is unavailable but the report-linked pathway remains. It does not estimate the diagnostic value of chest radiography outside that setting.

Evidence was interpreted in a fixed hierarchy. The report-preserving contrast and ICM are primary because they directly intervene on image content. The eight-seed distributions and paired tests quantify variation in the five fusion conditions. Calibration, per-class sensitivity, and paired decision counts are

secondary because they show whether a similar average score hides a change in confidence or error pattern. Gate extraction and Grad-CAM are supporting observations: they can describe the retained model but cannot replace an intervention.

This hierarchy guards against three common interpretation errors. First, the text-only and image-only controls are not treated as symmetric estimates of modality value because they use separate classifiers and only one retained run. Second, a gate near 0.5 is not described as equal causal contribution from image and text. Third, a non-significant difference from zero is not treated as equivalence. The operational equivalence claim is made only through TOST and only for the prespecified interval on the ICM scale.

The estimand is intentionally tied to Macro-F1. Let F_s^V denote the Macro-F1 of a full fusion condition at seed s , and let F^G denote the retained Gray-Square Macro-F1. The per-seed contrast is $F_s^V - F^G$, and ICM divides that contrast by F_s^V . This normalization makes the value easy to interpret relative to the full model, but it also means that ICM is metric-specific. A near-zero Macro-F1 ICM does not imply near-zero change for every class, threshold, or calibration measure; those quantities are examined separately.

The label-report relationship is part of the estimand, not a nuisance removed after analysis. CheXpert targets were derived from report language, and the model received the corresponding findings text. The audit therefore asks how a model behaves under that known information structure. It does not attempt to reconstruct an unobserved independent radiologist label or to reinterpret the report-derived target as image-only ground truth.

3.3 Preprocessing and architecture

The architecture exposes the contribution of each modality without changing the downstream classification target. Image and report features were encoded separately, projected to a common dimension, allowed to interact, and converted to class-specific logits. The gate therefore combines class-level evidence after feature extraction rather than deciding which raw input should be retained.

The **findings** field was tokenized with `emilyalsentzer/Bio_ClinicalBERT`, truncated or padded to 256 tokens, and randomly replaced by an empty string in 10% of training examples. Test and validation text was not masked. Images were converted to one channel, resized to 256×256 , randomly cropped to 224×224 during training,

and mapped from $[0, 255]$ to $[-1, 1]$. Training augmentation used horizontal flipping, rotation up to 10° , translation, and brightness/contrast jitter; the minority-class transform increased the rotation, translation, scale, shear, jitter, and blur ranges. Validation and test preprocessing was deterministic. Residual non-uniform effects of this mapping cannot be fully excluded; it is treated as a shared preprocessing constraint across all conditions [24].

The stronger minority-class augmentation was applied only during training. It was intended to increase variation for the rarer classes without changing validation or test inputs. No transformation used test labels or test-set statistics. All image-bearing fusion conditions used the same transform family and fixed split, so a cross-condition difference was not coupled to a separate evaluation pipeline.

The image branch used a TorchXRayVision DenseNet-121 with `densenet121-res224-mimic_ch` weights [24, 25]. All image-bearing fusion conditions shared the same $[-1, 1]$ intensity mapping. Any deviation from the native TorchXRayVision convention therefore acts as a shared preprocessing shift across conditions. ICM and condition contrasts characterize the retained preprocessing setting. The encoder’s convolutional stem and first dense block were frozen, and its output was projected to 512 dimensions through a linear layer, LayerNorm, Gaussian Error Linear Unit, and dropout (0.25). The text branch used Bio.ClinicalBERT with the first four transformer layers frozen and an analogous 512-dimensional projection [5, 6]. One projected image token was used as the query and one projected text token as the key/value in an eight-head attention block with residual LayerNorm. Because there is only one key, this operation cannot express word-region alignment; we interpret it as feature interaction, not causal attention.

The single image token and single text token create a compact fusion interface, but they also restrict what attention can represent. The attention output can mix projected modality features; it cannot resolve individual report tokens against spatial image regions. This boundary is carried into the interpretation of the qualitative saliency examples. The architecture permits a test of image sensitivity, but it does not support claims about phrase-level anatomical alignment.

Baseline CADQ combined image and text logits through a bounded learnable per-class gate [20]:

$$\begin{aligned}\alpha_c &= 0.20 + 0.60 \text{sigmoid}(u_c), \\ z_c &= \alpha_c z_c^{\text{img}} + (1 - \alpha_c) z_c^{\text{text}}.\end{aligned}\tag{1}$$

The initial logits $u = (-0.5, 0, 0.5)$ corresponded to image-gate values (0.426524, 0.500000, 0.573476) for Normal, Pneumonia, and Pneumothorax.

3.4 Experimental conditions and modality controls

The experimental design changes one aspect of fusion or regularization at a time while preserving the cohort and three-class endpoint. Baseline CADQ is the reference. LMH tests a learned text-prior correction, the noise-sensitivity (anti-consistency) condition tests response to perturbed images, exploratory GACR-B adds a grounding-aware penalty, and Adaptive-CADQ changes the gate from static to input-dependent. Table 2 records the intended contrast and the evidence boundary for each condition.

LMH added a five-fold out-of-fold term-frequency/inverse-document-frequency logistic text prior (5,000 unigram/bigram features, stop-word removal, balanced weights), temperature calibration, $\epsilon = 0.1$ smoothing, an input-dependent prior gate, and an entropy penalty of 0.05 annealed over three epochs. It was warm-started from CADQ and followed the product-of-experts rationale used for unimodal-bias reduction [18]. The noise-sensitivity (anti-consistency) condition perturbed the image with Gaussian noise ($\sigma = 0.25$). Its archived objective added $-D_{KL}(p_{\text{clean}} \| p_{\text{noisy}})$ to a minimized loss and therefore rewards divergence; throughout this manuscript it is treated as a noise-sensitivity (anti-consistency) condition, not a conventional consistency objective. Adaptive-CADQ replaced the static per-class gate with a sample-dependent scalar computed from image and text entropies by a $2 \rightarrow 16 \rightarrow 1$ multilayer perceptron. Exploratory GACR-B used an out-of-fold text baseline and a cosine penalty of 0.3 on examples for which that baseline misclassified, following the grounding-aware intervention of Sadanandan and Behzadan [12].

The image-only control used the image encoder with a 512-to-256 classifier; the text-only control used frozen Bio.ClinicalBERT with a 768-to-256 classifier. The Gray-Square control replaced every image by a constant intensity-128 tensor but preserved the report and multimodal architecture. Only one retained run was available for each control. Gray-Square is therefore a report-preserving image ablation, not an independently trained eight-seed text-only ensemble.

The three controls answer different questions. Image-only and text-only models describe the capacity of isolated pathways under their retained

training configurations. Gray-Square preserves the multimodal path while removing spatially varying image content. It is the primary comparator because the report, target, architecture, and evaluation cohort remain fixed. The single-run control coverage is reported explicitly so that its unmeasured run-to-run uncertainty is not confused with the eight-seed variation of the fusion conditions.

3.5 Training and outcomes

Training was held constant wherever the retained experiment allowed so that changes could be attributed to the tested condition rather than a separate optimization schedule. Model selection used validation Macro-F1 for the five multimodal conditions. Test metrics were computed after training and checkpoint selection; the held-out test set was not used to tune hyperparameters, select thresholds, or define the operational equivalence interval.

Models were trained for 10 epochs with AdamW and weight decay 10^{-4} [26]. Learning rates were 10^{-5} for the image and text backbones, 10^{-4} for fusion and LMH heads, and 5×10^{-5} for static gates. OneCycleLR used 10% warm-up [27]. Physical batch size was 8 with four gradient-accumulation steps (effective batch 32), mixed precision, and gradient scaling. The objective combined focal cross-entropy ($\gamma = 1.5$), label smoothing (0.05), and effective-number class weights ($\beta = 0.99$) [28]. Class-aware sampling replicated Pneumothorax examples up to fivefold and Pneumonia examples up to threefold. The highest validation Macro-F1 checkpoint was restored for multimodal conditions. The image-only control was retained as a descriptive capacity probe; its archived pipeline did not guarantee restoration of the best validation checkpoint. Inferential claims about modality reliance rest on the report-preserving Gray-Square intervention, not the image-only result.

Class-aware sampling changed the frequency with which examples were presented during optimization but did not change the held-out class distribution. The reported test metrics therefore describe the fixed 455-study test cohort rather than the sampling distribution used to construct mini-batches. Mixed precision and gradient accumulation were computational choices shared across conditions; they are recorded because small parameter movements, including gate drift, must be interpreted in the numerical setting in which checkpoints were saved.

Validation Macro-F1 aligned checkpoint selection with the primary performance scale while preserving the test set for final evaluation. The multimodal

conditions used the best retained validation checkpoint. The image-only control is kept as a capacity probe because its archived selection path did not provide the same guarantee. This asymmetry is why the image-only number is not used in ICM and why the report-preserving Gray-Square contrast anchors the main inference.

The primary diagnostic was

$$\text{ICM} = \frac{F1_{VLM} - F1_{Gray}}{F1_{VLM}}, \quad (2)$$

where positive values indicate better Macro-F1 with the real radiograph. The relative form was chosen to express the incremental change on the scale of the full model. An ICM near zero indicates that replacing the image did not materially change this endpoint; it does not prove that the image encoder is universally uninformative. The analysis used an operational equivalence interval of $[-0.01, 0.01]$. This project-defined margin is neither a validated clinical threshold nor a regulatory acceptance criterion.

ICM is signed. A positive value means the full model achieved higher Macro-F1 than the Gray-Square reference; a negative value means the retained reference was higher. The denominator expresses the change relative to the full-model score. We report the absolute Macro-F1 values beside ICM so that the normalized quantity cannot hide a small denominator or an otherwise poorly performing model. In this experiment the full-model scores are near 0.94, so absolute and relative differences are both readily interpretable.

Secondary outcomes were Macro-F1, accuracy, one-versus-rest Macro-AUROC, per-class F1 and sensitivity, ten-bin expected calibration error (ECE), multiclass Brier score, row-normalized confusion matrices, retained image-gate values, and qualitative Grad-CAM. ECE and Brier were recomputed from retained probability arrays when absent from the summary dictionary; missing values were never replaced with zero. Gate values were taken from the per-seed checkpoint export in `q1_audit_results.json`. The displayed gate figure uses all eight exported values and their equation-derived initialization references, rather than hard-coded aggregate values.

Macro-F1 was primary because the three classes were imbalanced and the rare Pneumothorax class should not be hidden by the larger Normal group. Accuracy and Macro-AUROC describe complementary aspects of prediction but cannot alone show whether a model uses the image. ECE and Brier score were included because two conditions can rank cases similarly while assigning materially different

Table 2. Experimental conditions and their audit roles.

Condition	Change from baseline	Question addressed	Interpretation boundary
Baseline CADQ	Static bounded per-class gate	Reference performance and image-replacement sensitivity	Gate value is not a causal contribution estimate.
LMH (PoE)	Learned text prior and product-of-experts correction	Does reducing the easy text route increase image contribution?	A lower score can reflect removal of target-linked report information.
Noise-sensitivity (anti-consistency)	Gaussian image perturbation with the retained negative-KL term	How does the archived objective respond to image noise?	The loss rewards divergence and is not conventional consistency training.
Exploratory GACR-B	Text-baseline-aware cosine penalty	Does a grounding-aware intervention alter contribution?	Added post hoc and interpreted as exploratory.
Adaptive-CADQ	Entropy-conditioned dynamic gate	Does input-dependent gating improve reliance over a static gate?	Dynamic weight variation does not by itself establish grounding.
Gray-Square	Constant image with report and multimodal path retained	Primary report-preserving image ablation	One retained run; comparator variance is not independently estimated.
Text-only and image-only	Separate unimodal classifiers	Descriptive capacity of each isolated pathway	Different retained control pipelines prevent a definitive modality ranking.

confidence. Per-class sensitivity and confusion matrices localize changes that an average metric can conceal.

Gate extraction was defined as a parameter audit rather than an attribution method. For every baseline seed, the saved checkpoint value was transformed by the same bounded sigmoid equation used during inference and compared with its class-specific initialization. We use the term *gate stagnation* for the observed pattern of negligible movement. The label describes parameter stability only; it does not establish that gradients were zero or identify the cause of that stability.

3.6 Statistical analysis and reproducibility

The statistical plan separates three questions. Paired tests assess whether a condition differs from baseline, equivalence tests assess whether ICM lies inside the operational near-zero interval, and bootstrap intervals summarize uncertainty across retained seeds. Subject-level tests were limited to matched seed-42 arrays because probability and prediction files for all comparable seeds were not uniformly available.

Seed-level differences used paired Wilcoxon signed-rank tests with Pratt zero handling and paired *t*-tests. ICM summaries included the median, standard deviation, Cohen’s *d*, and percentile bootstrap 95% confidence intervals from 2,000 re-

samples. Two one-sided tests (TOST) evaluated whether ICM lay within $[-0.01, 0.01]$ [22]. Because the same 455 studies recur at every seed, test subjects were not concatenated across seeds for confirmatory inference.

DeLong tests compared correlated Pneumothorax receiver-operating-characteristic curves within the retained seed-42 probability set [29]. McNemar tests compared paired classification decisions on the same 455 seed-42 studies [30]. Benjamini-Hochberg correction controlled the false-discovery rate within each test family [31]. A McNemar *p*-value of 1 denotes balanced discordance, not identical predictions; the two off-diagonal counts are therefore reported with the test result.

Python, NumPy, and PyTorch seeds were fixed and deterministic cuDNN settings were enabled. Run metadata, validation histories, checkpoints, predictions, probabilities, and configurations were retained. The fixed merge archive resolved duplicate keys by the first-seen value, but the earlier workflow did not preserve a deterministic source-priority rule and content-hash manifest. Results are consequently tied to this archive snapshot. Reporting was audited against CLAIM and TRIPOD+AI principles [32, 33]; checklist summaries and a provenance manifest are included in the Supplementary material.

Analyses were performed on fixed retained outputs rather than by pooling predictions from sep-

arate seeds into a larger artificial sample. This preserves the trained model as the replication unit for seed-level comparisons. It also avoids confidence intervals that would appear too narrow because the same 455 patients recur under every seed. All inferential results are reported with the test family, unit of analysis, and multiplicity adjustment needed to interpret them.

4 Results

Results are presented in the order needed to evaluate the modality claim. We first describe predictive performance across the five eight-seed conditions, then examine paired tests and calibration. The report-preserving Gray-Square comparison provides the primary evidence for incremental image contribution. Gate extraction and qualitative saliency are reported last because they support interpretation but do not independently establish modality use.

4.1 Analysis set, discrimination, and calibration

The fixed archive contained 40 condition-seed records: five conditions with eight seeds each, evaluated on the same 455 held-out studies. Baseline CADQ achieved Macro-F1 0.9397 ± 0.0061 , accuracy 0.9607 ± 0.0032 , and Macro-AUROC 0.9974 ± 0.0004 (Table 3). LMH had the highest mean Macro-F1 (0.9435 ± 0.0093) but lower and more variable AUROC (0.9849 ± 0.0113). The noise-sensitivity (anti-consistency) condition had the lowest Macro-F1 (0.9346 ± 0.0075). Exploratory GACR-B reached 0.9417 ± 0.0059 , and Adaptive-CADQ remained close to baseline at 0.9388 ± 0.0072 .

The narrow range of mean Macro-F1 values shows that the five optimization strategies reached similar aggregate endpoints. The ranking alone is not sufficient to identify an improvement: the largest mean belongs to LMH, but its seed variability is wider and its calibration and AUROC move in the opposite direction. The baseline therefore remains the reference for the modality audit rather than being replaced by the condition with the highest unadjusted mean.

The multiplicity-adjusted tests did not identify a Macro-F1 improvement over baseline. At seed 42, LMH and the noise-sensitivity (anti-consistency) condition produced measurable AUROC degradation even though their absolute AUROC values remained high. This combination illustrates why ceiling-level discrimination can hide a directional

loss. The McNemar results add a separate view: they test whether paired errors changed direction, not whether two prediction vectors are identical.

Baseline per-class F1 was 0.9755 ± 0.0032 for Normal, 0.9624 ± 0.0055 for Pneumonia, and 0.8810 ± 0.0174 for Pneumothorax. LMH increased Pneumothorax F1 to 0.9003 ± 0.0191 but reduced Pneumonia F1 to 0.9575 ± 0.0077 . The noise-sensitivity (anti-consistency) condition had the widest Pneumothorax variation (0.8753 ± 0.0237). Adaptive-CADQ showed the most stable Pneumonia F1 (0.9616 ± 0.0029). The distributions and retained controls are shown in Fig. 2. The control panel is not an accuracy ranking across equally replicated systems: each control represents one retained run, whereas each fusion distribution contains eight seeds.

No condition improved Macro-F1 over baseline after Benjamini-Hochberg correction (Table 4). The largest unadjusted Macro-F1 signal was the exploratory GACR-B contrast ($\Delta = 0.0020$, Wilcoxon $p = 0.0547$; adjusted $p = 0.2188$). LMH and the noise-sensitivity (anti-consistency) condition had lower seed-42 Pneumothorax AUROC than baseline; their significant DeLong results indicate degradation, not improvement. Adaptive-CADQ showed no AUROC difference.

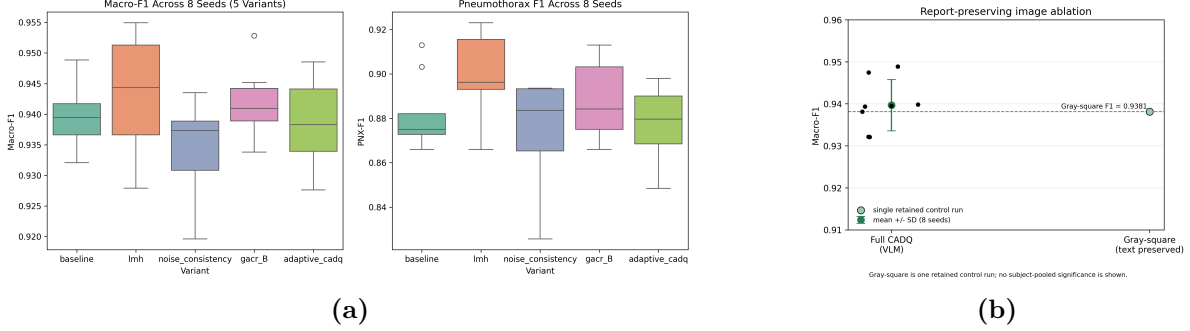
Table 4 uses eight paired seed-level Macro-F1 values for the Wilcoxon comparisons. DeLong and McNemar use the retained seed-42 probability and prediction arrays for the same 455 studies. Parenthetical values are Benjamini-Hochberg adjusted within each test family, and n_{01}/n_{10} are the two McNemar discordant counts.

The balanced discordance for exploratory GACR-B and Adaptive-CADQ is informative. Exploratory GACR-B disagreed with baseline on 17 studies (9 versus 8 across the two off-diagonal directions), and Adaptive-CADQ on 29 (15 versus 14). McNemar $p = 1.000$ indicates no evidence of directional asymmetry in these paired errors; it does not indicate byte-identical predictions. The confusion matrices and receiver-operating-characteristic curves in Fig. 3 are descriptive views of the same repeated test cohort. The confusion matrices are row-normalized and aggregated over seeds. The ROC bands summarize multi-seed variation, whereas DeLong inference uses matched seed-42 arrays and was not performed on subjects pooled across seeds. Significant DeLong results for LMH, the noise-sensitivity (anti-consistency) condition, and exploratory GACR-B indicate lower Pneumothorax AUROC than baseline rather than improvement.

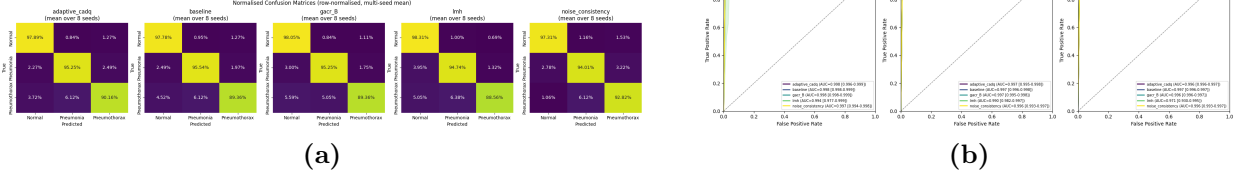
Calibration separated some conditions that had

Table 3. Performance and calibration across eight seeds.

Condition	Macro-F1	Accuracy	Macro-AUROC	ECE	Brier
Baseline CADQ	0.9397 ± 0.0061	0.9607 ± 0.0032	0.9974 ± 0.0004	0.0237 ± 0.0048	0.0522 ± 0.0040
LMH (PoE)	0.9435 ± 0.0093	0.9596 ± 0.0070	0.9849 ± 0.0113	0.0621 ± 0.0160	0.0685 ± 0.0213
Noise-sensitivity (anti-consistency)	0.9346 ± 0.0075	0.9560 ± 0.0039	0.9960 ± 0.0013	0.0347 ± 0.0137	0.0655 ± 0.0130
Exploratory GACR-B	0.9417 ± 0.0059	0.9610 ± 0.0042	0.9972 ± 0.0005	0.0245 ± 0.0023	0.0540 ± 0.0038
Adaptive-CADQ	0.9388 ± 0.0072	0.9610 ± 0.0044	0.9972 ± 0.0008	0.0241 ± 0.0070	0.0527 ± 0.0028

**Fig. 2.** Fusion-condition F1 distributions and retained modality controls.**Table 4.** Paired comparisons with the baseline condition.

Condition	Wilcoxon Macro-F1 p (BH)	DeLong PNX-AUROC p (BH)	McNemar n_{01}/n_{10} ; p (BH)
LMH (PoE)	0.2500 (0.3333)	2.9×10^{-6} (1.2×10^{-5})	27/31; 0.6936 (1.0000)
Noise-sensitivity (anti-consistency)	0.1953 (0.3333)	0.0022 (0.0044)	32/49; 0.0754 (0.3018)
Exploratory GACR-B	0.0547 (0.2188)	0.0315 (0.0420)	9/8; 1.0000 (1.0000)
Adaptive-CADQ	0.8438 (0.8438)	0.7304 (0.7304)	15/14; 1.0000 (1.0000)

**Fig. 3.** Row-normalized confusion matrices and class-specific ROC curves.

similar Macro-F1. Baseline ECE was 0.0237 ± 0.0048 and Brier score 0.0522 ± 0.0040 . LMH increased ECE to 0.0621 ± 0.0160 (2.6-fold) and Brier score to 0.0685 ± 0.0213 ; the adjusted seed-level p -values were 0.0313 and 0.0156. The noise-sensitivity (anti-consistency) condition increased Brier score to 0.0655 ± 0.0130 (adjusted $p = 0.0156$), whereas its ECE change was inconclusive. Exploratory GACR-B and Adaptive-CADQ remained close to baseline. Figure 4 shows the baseline reliability diagram and seed-level ECE distributions.

The reliability panel is descriptive and uses confidence bins to compare observed correctness with predicted confidence. The seed-level ECE panel shows the dispersion hidden by a single mean. Together they indicate that LMH’s small Macro-F1 difference did not represent a uniformly better model: confidence quality deteriorated while classification

performance remained close to the baseline range.

4.2 Report-preserving controls and ICM

The retained image-only control achieved Macro-F1 0.4611, accuracy 0.5670, Macro-AUROC 0.6728, ECE 0.0993, and Brier score 0.6142. The text-only control achieved 0.8555, 0.9077, 0.9820, 0.0203, and 0.1289, respectively. The Gray-Square control achieved Macro-F1 0.9381, accuracy 0.9604, Macro-AUROC 0.9960, ECE 0.0369, and Brier score 0.0589. Its Macro-F1 was only 0.0016 below the baseline eight-seed mean. Because these were single retained runs and the image-only pipeline did not guarantee restoration of the best validation checkpoint, they support an audit of the archived system but not a definitive ranking of modality capacity.

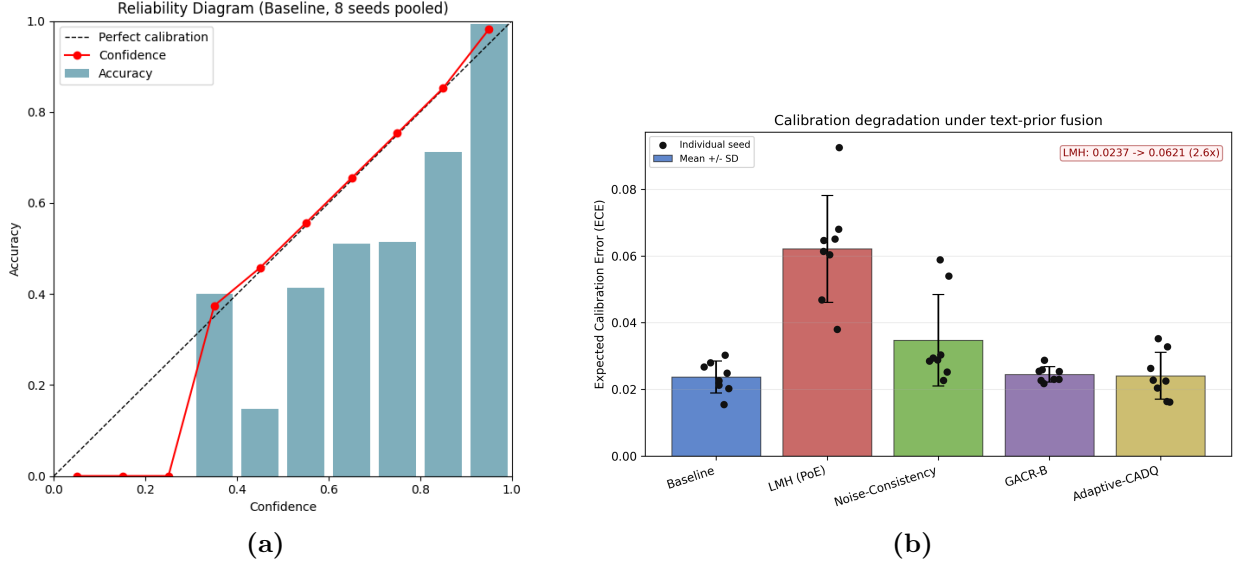


Fig. 4. Baseline reliability and seed-level expected calibration error.

Baseline ICM was 0.0016 ± 0.0065 with median 0.0014 and bootstrap 95% confidence interval $[-0.0026, 0.0061]$ (Table 5). The signed-rank test did not reject zero ($p = 0.4219$), while TOST supported equivalence inside the operational interval ($p = 0.0040$). LMH had ICM 0.0056 ± 0.0098 and did not meet the equivalence criterion. The noise-sensitivity (anti-consistency) condition had a negative mean ICM (-0.0039 ± 0.0081), whereas exploratory GACR-B and Adaptive-CADQ had small positive values. None showed a robust positive contribution relative to the Gray-Square reference.

The baseline estimate is central because its confidence interval straddles zero while remaining within the predeclared operational bounds. The Wilcoxon result alone would only show that the retained seeds do not establish a difference from zero. The TOST result supplies the additional, bounded claim that the effect is compatible with the chosen near-zero interval. This is an operational statement about Macro-F1 in this cohort, not proof of exact equality and not a clinical non-inferiority margin.

The confidence intervals in Table 5 are percentile bootstrap intervals from 2,000 seed resamples. The Wilcoxon column tests ICM against zero, whereas TOST asks whether it lies inside the operational interval. These tests answer different questions and are therefore reported together.

The seed pairing, equivalence interval, and retained gate values are shown in Fig. 5. The Gray-Square value was reused for all eight ICM calculations; its run-to-run uncertainty is therefore absent from the displayed intervals. The equivalence result should be interpreted as a property of the retained

comparator, not an estimate from two independently replicated arms.

4.3 Gate behaviour and qualitative inspection

The retained baseline gate means were 0.42624 for Normal, 0.49963 for Pneumonia, and 0.57336 for Pneumothorax. The overall mean was 0.49975 compared with an initialization mean of 0.50000, and the largest per-seed class-wise deviation from the equation-derived initialization was 0.000499. The gate did not collapse toward its lower bound of 0.20, nor did the retained values show a meaningful shift toward greater image reliance. We call this descriptive pattern *gate stagnation*; it is evidence that this parameter changed little, not proof that its gradient was exactly zero or that the image branch was disconnected. Per-seed values appear in Supplementary Table S1.

The class-specific means track the deliberately different initial values rather than converging on a common boundary. This pattern explains why a nominal gate near 0.5 cannot be read as evidence that half of the decision came from the image. The gate equation allocates weight between logits, but the represented features can be redundant and the parameter can remain almost unchanged. The behavioural Gray-Square comparison is therefore the stronger source of evidence about incremental image information.

The three retained Grad-CAM examples showed diffuse activation in Baseline seed 42 (Supplementary Fig. S1). They were not compared with localization annotations, deletion curves, or clinician as-

Table 5. Seed-level Image Contribution Metric statistics.

Condition	Mean	SD	Median	95% CI	Wilcoxon p	TOST p	Cohen's d
Baseline	0.0016	0.0065	0.0014	$[-0.0026, 0.0061]$	0.4219	0.0040	0.247
LMH (PoE)	0.0056	0.0098	0.0066	$[-0.0011, 0.0118]$	0.1953	0.1223	0.566
Noise-sensitivity (anti-consistency)	-0.0039	0.0081	-0.0009	$[-0.0097, 0.0009]$	0.3828	0.0345	-0.476
Exploratory GACR-B	0.0038	0.0062	0.0030	$[-0.0002, 0.0081]$	0.1953	0.0122	0.607
Adaptive-CADQ	0.0007	0.0077	0.0002	$[-0.0043, 0.0057]$	0.8438	0.0054	0.089

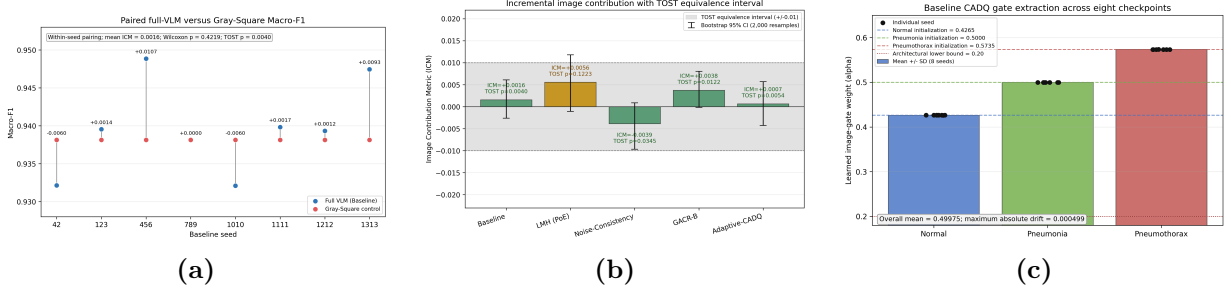


Fig. 5. Paired image ablation, ICM equivalence, and retained baseline gates.

essment. They therefore provide only a qualitative sanity check and cannot establish visual grounding or its absence.

4.4 Synthesis across evidence sources

The result set is internally consistent when each analysis is assigned its proper scope. Baseline discrimination is high, but the report-preserving control retains nearly the same Macro-F1. The baseline ICM confidence interval crosses zero and fits inside the operational equivalence range under TOST. The static gate remains close to initialization, and the qualitative saliency examples do not supply independent localization evidence. None of these observations alone proves the absence of visual information; together they show that the evaluated Macro-F1 provides little evidence of incremental image use in the retained setting.

The intervention results also explain why the cross-variant table should not be read as a simple ranking. LMH has the largest mean Macro-F1, yet its AUROC and calibration are worse than baseline. Exploratory GACR-B has the largest small positive ICM among the tested variants, but the adjusted Macro-F1 comparison does not establish improvement. Adaptive-CADQ retains baseline-like discrimination and calibration without increasing ICM. The noise-sensitivity (anti-consistency) condition changes the objective but does not produce a positive contribution estimate. No condition satisfies the combined requirement of improved incremental image contribution without an offsetting loss in the reported evidence.

Class-specific results place the aggregate finding

in context. Pneumothorax has the smallest test count and the widest variation, so small numerical changes in its F1 require more caution than equally sized changes in the Normal class. Seed-42 sensitivity intervals overlap across conditions, and the descriptive confusion matrices do not reveal a new, stable rare-class advantage. This does not eliminate the possibility of a clinically important subgroup effect; it shows that such an effect is not established by the retained cohort and outputs.

Finally, the evidence distinguishes reproducibility from breadth. Forty condition-seed records support paired seed-level summaries for the five fusion conditions. The controls, DeLong analysis, and McNemar analysis have narrower retained coverage. Their value lies in answering specific audit questions, not in implying that every component has the same replication depth. This coverage is repeated in the supplementary provenance manifest so that readers can trace each numerical claim to the available output type.

5 Discussion

This section interprets the findings at three levels. It first examines what the Gray-Square result establishes about the evaluated construct, then relates the fusion and calibration results to prior modality-bias work, and finally sets out the conditions required for broader biomedical or clinical use. The discussion keeps behavioural evidence separate from possible optimization mechanisms so that a descriptive audit is not presented as a causal account of training dynamics.

5.1 Principal findings and interpretation

The central result is a mismatch between high predictive performance and incremental image evidence. Baseline Macro-F1 approached 0.94, yet a constant gray image paired with the same report retained Macro-F1 0.9381. Baseline ICM was close to zero, its signed-rank test was non-significant, and TOST supported equivalence within the project-defined interval. None of the four alternative fusion or text-debiasing conditions produced a robust ICM increase. The LMH text-prior intervention had the highest mean Macro-F1 but lower AUROC and much worse calibration. The retained baseline gates stayed close to initialization. Together, these findings show that aggregate performance, a non-collapsed gate, and visually plausible activation maps can coexist with negligible performance change after image replacement.

This combination is more informative than any component alone. High baseline performance shows that the task is predictable under the available inputs. The Gray-Square contrast asks whether image content adds to that prediction. TOST places the observed difference within an operational interval, and gate extraction checks whether the fusion parameter moved during training. The four steps narrow the interpretation from “the model performs well” to “the model performs well even when image content is removed under this retained report-linked setting.”

5.2 Construct validity and modality inertia

The most direct explanation is construct overlap. MIMIC-CXR pairs each image with a report, CheXpert uses report language to generate the target, and the model receives the report’s **findings** field [1, 13]. The Gray-Square control preserves this high-information route. It does not show that the CheXpert labels are clinically inaccurate, that every prediction copied a particular phrase, or that radiographs lack diagnostic value. It shows that the evaluated endpoint does not isolate independent image-based recognition. The claim supported by the data is therefore narrow: under this report-linked setup and archived implementation, the measured incremental image contribution was negligible.

The distinction is one of construct validity. If the intended construct is independent visual recognition, a target generated from the same report given to the model does not isolate it. If the intended construct

is report-linked classification, the task remains internally coherent, but the meaning of the score changes. The same numerical result can therefore be appropriate for one use and misleading for another. A clear target-provenance statement is needed before performance is compared with image-only or independently labelled systems.

We use *modality inertia* to describe the combined observation that the output changes little under report-preserving image replacement while the static gate remains near its starting value. This is a useful diagnostic label, but the current experiment does not identify a unique mechanism. Redundant features, optimization dynamics, precision, normalization mismatch, or weak image gradients could each contribute. The gate result should not be promoted from a parameter-stability observation to proof of causal gradient starvation without gradient traces or controlled retraining.

The term also avoids a false choice between gate collapse and successful fusion. A gate can stay comfortably inside its bounds while the output remains insensitive to the image. That outcome is possible when the two pathways encode overlapping evidence or when one pathway already predicts the report-derived target. Gate values should therefore be interpreted together with ablations, not converted directly into percentages of causal contribution.

5.3 Interpretation of the tested conditions

The debiasing results clarify why endpoint design matters. Product-of-experts methods are intended to suppress an easy language prior. Here, however, report language helped generate the target. Down-weighting this pathway can degrade AUROC or calibration without creating new image information, as observed for LMH. The noise-sensitivity (anti-consistency) condition cannot support its intended conventional-consistency interpretation because its archived negative-KL formulation rewards divergence. Adaptive-CADQ changed the gate form but not the endpoint. Exploratory GACR-B, following the grounding-aware proposal of Sadanandan and Behzadan [12], achieved a small ICM but not a robust Macro-F1 improvement. The experiment did not measure whether its $\lambda = 0.3$ penalty generated insufficient gradient; that mechanism requires a dedicated ablation and is not inferred from the null performance result.

These findings do not show that debiasing is generally ineffective. They show that each intervention is evaluated against the information structure of

its target. Product-of-experts methods can be useful when a language prior is a spurious correlate; noise-based objectives can be useful when their loss encourages stable predictions; adaptive gates can respond when modality uncertainty is informative; and grounding-aware penalties can matter when the reference standard rewards the intended evidence. In this experiment, none of those changes established a reliable gain over the report-preserving control. The relevant conclusion concerns the fit between method and benchmark, not a universal ranking of algorithms.

The corrected McNemar interpretation is also important. Exploratory GACR-B and Adaptive-CADQ had $p = 1.000$, but both had non-zero discordant predictions. The off-diagonal counts were nearly balanced, so McNemar detected no directional change in error rates. This supports a conclusion of similar aggregate paired behaviour, not byte-identical decision boundaries. Reporting the contingency counts prevents a non-significant symmetry test from being mistaken for an identity test.

Calibration provides a second reason to avoid ranking conditions from Macro-F1 alone. LMH reached the highest unadjusted mean Macro-F1 but increased ECE by a factor of 2.6 and increased Brier score. A system can preserve its predicted class while changing probability quality enough to affect threshold-based use. For biomedical evaluation, a debiasing objective should therefore be judged by discrimination, calibration, and modality contribution together.

5.4 Implications for biomedical informatics

The practical implication is a change in evaluation sequence. Target provenance should be mapped before a multimodal architecture is trained. Modality controls should then be matched to the intended claim, and only after that should aggregate performance be interpreted. This order helps separate a useful report-conditioned predictor from a claim of independent visual reasoning.

Multimodal evaluation should make modality contribution explicit. A minimum audit should document how targets were produced, whether any input shares that source, what happens when each modality is replaced or permuted, whether controls are matched across seeds, and how calibration changes. The ablation must preserve all other inputs and the evaluation cohort. Where the scientific claim is that an image contributes, an independent or image-derived reference standard is preferable to a target

extracted from the accompanying report. Counterfactual text tests, temporally separated note fields, and cross-patient report swaps can further reveal which source carries the prediction.

Prior studies support this sequence from different directions. Image-text learning establishes the value of report supervision [2–4]; context and image-removal studies show that apparent multimodal performance can persist without the intended visual evidence [10, 11]. The audit taxonomy of Sadanandan and Behzadan provides a broader framework for interpreting that behaviour [12]. The present study adds a seed-level relative measure, an operational equivalence test, and retained-gate inspection. These forms of evidence are complementary rather than interchangeable.

ICM is one compact summary of such an intervention. Its advantages are an interpretable direction and a model-relative scale, but it inherits every limitation of the chosen performance metric and comparator. Macro-F1 may hide class-specific harm; a single Gray-Square run understates control uncertainty; and the $[-0.01, 0.01]$ interval has not been clinically validated. ICM should therefore be reported with absolute performance, per-class outcomes, calibration, confidence intervals, and a description of the control. Validation on independent-label tasks is needed before comparing ICM across datasets.

Gate and saliency analyses should remain secondary. A gate is shaped by parameterization and can be unidentifiable when modality signals are redundant. The one-key attention block used here cannot align words to image regions, and three Grad-CAM examples lack quantitative localization evidence. A stronger future design would use token- and region-level interactions, paired modality interventions, gradient and gate trajectories, concept localization, and reader-validated explanations. Even then, behavioural change under input intervention remains the more direct evidence of contribution.

Reporting should also distinguish primary and supporting evidence. A compact minimum set would include the full-condition metric, a matched ablation, the absolute and relative difference, uncertainty over independent training runs, class-specific outcomes, and calibration. Architecture parameters and explanations can be added when their scope is explicit. CLAIM and TRIPOD+AI provide useful structures for data, modelling, and validation, while a modality-provenance table and intervention result make the specific information flow easier to audit [32, 33].

For regulatory-facing evidence, the central les-

son is traceability rather than a proposed mandatory threshold. Good machine-learning-practice and real-world- evidence principles emphasize representative evaluation, transparent data provenance, uncertainty, and evidence aligned with intended use [34, 35]. A multimodal device dossier should therefore state whether labels and inputs share provenance and include matched ablations when modality use is part of the intended claim. ICM and gate inspection can support that evidence package, but neither substitutes for external validation, clinical utility, subgroup performance, or human-factors assessment.

5.5 Requirements for broader validation

A broader study should first break the target-input dependence. The preferred reference would be an image-derived label established independently of the report text given to the model, with adjudication rules and uncertainty handling specified before evaluation. If report-conditioned prediction is the intended task, that use should be stated directly and compared with a text-only system. The key requirement is alignment between the claimed capability and the source of the reference standard.

Control replication is the next priority. Full VLM, Gray-Square, text-only, and image-only conditions should be trained under the same seed schedule and checkpoint rule. Image replacement can be complemented by image permutation, cross-patient replacement, region masking, and report perturbation. Those interventions should be selected according to the question: replacement tests whether image content matters at all, permutation tests whether image-report pairing matters, and localized masking tests whether specific regions affect the output. A single control cannot answer all three.

The statistical design should specify the replication unit and minimum relevant effect before model fitting. When several seeds are paired to the same control, both sides of the comparison should carry replicated uncertainty. Subject-level ROC and error tests should use matched predictions from the same seed or an explicit hierarchical model rather than concatenating repeated patients. Equivalence margins should be justified for the chosen metric and intended use; the operational $[-0.01, 0.01]$ interval in this study is a transparent project definition, not a transferable clinical standard.

Evaluation should extend beyond one aggregate score. Class-specific sensitivity and precision are needed when prevalence and clinical consequences

differ. Calibration should be assessed before and after any probability adjustment, with thresholds selected on validation data. Subgroup analyses should be planned for demographic, acquisition, and institutional factors that are available and ethically appropriate. External evaluation should preserve the target definition or explicitly explain how a different label process changes the estimand.

Mechanistic claims require their own measurements. To test whether weak gradient signal explains gate stagnation, an experiment would need gate and gradient trajectories, controlled initialization, full-precision checks, and training under an independently labelled endpoint. To assess spatial grounding, qualitative heatmaps should be supplemented by expert localization, deletion or insertion tests, and reader assessment. These measurements would connect internal model behaviour to the intervention result without treating a single visualization as causal evidence.

Finally, the evidence chain should be recoverable. A release should map every condition to the source notebook, dataset and split hashes, software environment, checkpoint rule, seed, probability array, prediction file, and figure-generation command. This form of provenance is useful even when raw clinical data cannot be redistributed. It allows a reviewer to distinguish a reproducible numerical result from an interpretation that depends on an unstated processing choice.

5.6 Study boundaries

This audit has substantial limitations. It used 2,953 studies from one database, one labeler, three classes, and only 47 Pneumothorax test studies. The uncertain-to-negative mapping, class caps, prevalence, and inherited split may affect the results. No external cohort, radiologist-adjudicated reference, prospective workflow, reader study, subgroup analysis, decision-curve analysis, or outcome evaluation was available. Generalization to other reports, institutions, devices, and label processes is unknown.

Implementation limitations constrain causal interpretation. Residual non-uniform effects of the $[-1, 1]$ mapping cannot be fully excluded; it is treated as a shared preprocessing constraint across all conditions [24]. The noise-sensitivity (anti-consistency) condition used a negative-KL formulation that rewards divergence. The attention block used one token per modality. Image-only checkpoint restoration did not guarantee the best validation checkpoint, so that control is descriptive. The affected conditions are interpreted as probes of the

archived system’s behaviour under a report-linked endpoint. ICM and condition contrasts are the primary evidence within this setting; although shared preprocessing makes condition-specific effects unlikely, residual non-uniform effects cannot be fully excluded.

Control and statistical asymmetries also matter. The same single Gray-Square run was reused to construct eight ICM values, so comparator uncertainty was not propagated. Eight seeds provide a small replication sample. DeLong and McNemar results were computed only from the retained seed-42 arrays, whereas seed-aggregated curves are descriptive. Adaptive-CADQ’s eight-seed extension and exploratory GACR-B were post hoc. No a priori sample-size calculation established power for a clinically meaningful ICM. Future studies should train every control at every seed, preregister the equivalence margin and estimand, and use a larger independent-label cohort.

The ICM formulation has additional measurement boundaries. Because it is normalized by the full-model Macro-F1, it can become unstable for poorly performing models and should always be reported beside the absolute difference. The constant gray tensor removes anatomical information but can also create an input distribution that differs from natural radiographs. A cross-patient image swap or matched permutation control would test pairing dependence with more natural image statistics. The current study did not retain such a replicated control, so its conclusion is specific to the Gray-Square intervention.

The supporting analyses are narrower than the primary endpoint. Gate values were available for baseline checkpoints but not as full trajectories for every condition. The small numerical drift is compatible with stagnation, yet the retained precision limits fine-grained interpretation. Grad-CAM was available for three baseline seed-42 examples without expert spatial annotations. These examples can reveal obvious qualitative behaviour, but they do not measure localization accuracy or the causal effect of a highlighted region.

Finally, the result archive was assembled from multiple notebook outputs. The merge retained first-seen duplicates without a documented source-priority rule or content hashes, and some probabilities had to be recovered from a companion pickle. A confirmatory release should include a locked environment, dataset and split hashes, notebook and checkpoint hashes, deterministic merge rules, full probability arrays, and a machine-readable provenance manifest. The present manifest is descriptive

and identifies rather than resolves these issues.

6 Conclusion

In this MIMIC-CXR-derived task, replacing the radiograph with a constant image preserved nearly all baseline Macro-F1. Baseline ICM was near zero and equivalent within the operational interval, the tested interventions did not demonstrate a robust increase in incremental image contribution, and retained static gates remained near initialization. These findings expose a benchmark-validity risk created when labels and text inputs share report provenance. They do not show that radiographs or medical VLMs are generally uninformative. Independent labels, implementation validation, matched multi-seed controls, provenance locking, and external validation are required. ICM is best used as one component of a broader modality-reliance audit.

Ethics approval and consent to participate

MIMIC-CXR is de-identified and was accessed through official PhysioNet credentialing under its Data Use Agreement. No participants were recruited, no direct identifiers were accessed, and consent was not applicable. The source resource was released under the originating institutional review-board approvals; this secondary computational analysis introduced no intervention or participant contact.

Data availability

Raw MIMIC-CXR images and reports remain under PhysioNet access controls and are not redistributed. Non-identifying summaries, code, manuscript source, and figure-generation materials are available at <https://github.com/i-mTanvir/MediVision>. Cohort identifiers can be shared only in a manner consistent with the PhysioNet Data Use Agreement.

Code availability

The analysis notebooks, LaTeX source, figure scripts, and environment records are available at <https://github.com/i-mTanvir/MediVision>. A permanent archival DOI will be added if a versioned release is deposited. Credentials, raw reports, and restricted identifiers are excluded.

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Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced this work.

Author contributions

Nafis Al Rahman and Tanvir Mahmud contributed equally to conceptualization, methodology, investigation, software, data curation, formal analysis, validation, visualization, and writing. Both authors reviewed and approved the manuscript and accept responsibility for its content.

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Declaration of generative AI and AI-assisted technologies in the writing process

Generative-AI tools were not used to generate study data, execute experiments, create scientific results, or make analytical decisions. Any language or formatting assistance used during manuscript preparation was reviewed and edited by both authors, who take full responsibility for the final content.

S Supplementary material

S.1 Per-seed gate values

Table S1 reports the checkpoint value for every baseline seed; the reference initialization is (0.426524, 0.500000, 0.573476). Maximum drift is the largest absolute class-wise difference from initialization for that seed.

Table S1. Baseline gate values from retained checkpoints.

Seed	Normal α	Pneumonia α	Pneumothorax α	Max drift
42	0.426270	0.499512	0.573242	0.000488
123	0.426270	0.499512	0.573730	0.000488
456	0.426025	0.499756	0.573242	0.000499
789	0.426270	0.499756	0.573242	0.000255
1010	0.426270	0.499756	0.573242	0.000255
1111	0.426270	0.499512	0.573242	0.000488
1212	0.426270	0.499756	0.573730	0.000255
1313	0.426270	0.499512	0.573242	0.000488

S.2 Training configuration

Table S2. Training configuration for retained experiments.

Setting	Value
Image encoder	TorchXRayVision DenseNet-121, res224 MIMIC weights
Text encoder	Bio.ClinicalBERT; first four transformer layers frozen
Optimizer	AdamW; weight decay 10^{-4}
Learning rates	10^{-5} image/text; 10^{-4} heads; 5×10^{-5} static gates
Schedule	OneCycleLR with 10% warm-up; 10 epochs
Batching	Physical batch 8; four accumulation steps; effective batch 32
Objective	Focal cross-entropy ($\gamma = 1.5$), label smoothing 0.05, effective-number $\beta = 0.99$
Precision	FP16 mixed precision with gradient scaling
Gate initialization	$u = (-0.5, 0, 0.5)$; $\alpha = (0.426524, 0.500000, 0.573476)$
Augmentation	Crop, flip, rotation, translation, intensity jitter; minority-class transform
Seeds	42, 123, 456, 789, 1010, 1111, 1212, 1313
Hardware	Kaggle NVIDIA T4 two-GPU environment

S.3 Reporting-checklist summaries

The summaries below preserve the counts from the project-level reporting audits. Full item-level responses remain in the repository.

Table S3. CLAIM 2024 reporting summary.

Status	Count	Summary
Yes	13	Study type, source, outcome, model description, metrics, comparisons, limitations, and seed-level analysis were reported.
Partial	10	External validation, comparator breadth, ablation coverage, and reproducibility were limited by scope and controlled data access.
No	4	Cross-validation, external state-of-the-art baselines, pretraining-overlap audit, and complete credentialing documentation were not included.

Table S4. TRIPOD+AI reporting summary.

Status	Count	Summary
Yes	9	The title, data source, participants, model, statistical tests, limitations, registration history, and availability statements were addressed.
Partial	18	Abstract detail, hyperparameters, comparison breadth, reproducibility, and ethics reporting were only partly addressed.
No	14	External validation, decision-curve analysis, subgroup/fairness analysis, missing-data assessment, calibration transfer, and a model card were absent.

S.4 Provenance manifest

The manifest identifies retained outputs and constraints for each condition.

Table S5. Condition-level provenance manifest.

Condition	Seed coverage	Retained outputs	Known constraint
Baseline CADQ	8 seeds	Metrics, predictions, probabilities, gates	Shared $[-1, 1]$ mapping; residual non-uniform effects not excluded
LMH (PoE)	8 seeds	Metrics, predictions, probabilities	Warm-started; calibration degraded
Noise-sensitivity (anti-consistency)	8 seeds	Metrics, predictions, probabilities	Negative-KL formulation rewards divergence
Exploratory GACR-B	8 seeds	Metrics, predictions, recovered probabilities	Added post hoc; merge provenance was not hash-locked
Adaptive-CADQ	8 seeds	Metrics, predictions, probabilities	Extended from three to eight seeds post hoc
Controls	One retained run each	Metrics and selected arrays	Gray-Square uncertainty not replicated; image-only best-checkpoint restoration was not uniform

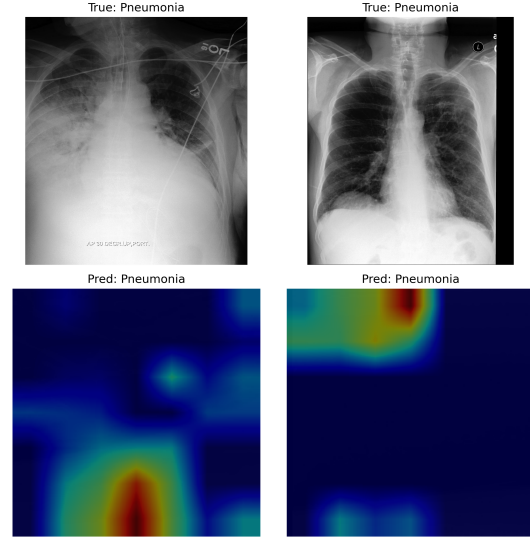
S.5 Seed-42 sensitivity intervals

Table S6 uses the seed-42 test counts: 240 Normal, 168 Pneumonia, and 47 Pneumothorax studies.

Table S6. Seed-42 sensitivity with Wilson 95% confidence intervals.

Condition	Normal	Pneumonia	Pneumothorax
Baseline	0.975 [0.946, 0.988]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
LMH (PoE)	0.962 [0.929, 0.980]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
Noise-sensitivity (anti-consistency)	0.962 [0.929, 0.980]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
Exploratory GACR-B	0.975 [0.946, 0.988]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
Adaptive-CADQ	0.979 [0.952, 0.991]	0.953 [0.910, 0.976]	0.894 [0.774, 0.954]

S.6 Qualitative saliency examples

**Fig. S1.** Qualitative baseline Grad-CAM examples for seed 42.

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